

Using the Spreadsheet in Engineering Calculations

F. Okyar

ME 101 Introduction to Mechanical Engineering



Department of Mechanical Engineering
Yeditepe University

Definitions

A spreadsheet application is a computer program such as MS Excel, LibreOffice Calc, or Google Sheets.

It has a number of built in features and tools, such as functions, formulas, charts, and data analysis tools allowing us to work with large amounts of data.*

*http://spreadsheets.about.com/od/s/g/spreadsheet_def.htm

Purposes (1)

0 Manage large amounts of raw data



EXPERIMENTAL:
Sensors, transducers, etc

Introduction to Mechanical Engineering

	A	B	C	D	E	F
1	Süre	Uzama	Gerilim 1	Yük		
2	(sec)	(mm)	(mm/mm)	(N)		
1845	377,2	24,5	0,011	10427,53		
1846	377,4	24,5	0,011	10373,7		
1847	377,6	24,5	0,011	10320,67		
1848	377,8	24,5	0,011	10258,01		
1849	378	24,6	0,011	10201,07		
1850	378,2	24,6	0,011	10143,59		
1851	378,4	24,6	0,011	10083,28		
1852	378,6	24,6	0,011	10020,11		
1853	378,8	24,6	0,011	9959,081		
1854	379	24,6	0,011	9897,514		
1855	379,2	24,6	0,011	9829,208		
1856	379,4	24,7	0,011	9764,295		
1857	379,6	24,7	0,011	9695,724		
1858	379,8	24,7	0,011	9622,309		
1859	380	24,7	0,011	9536,194		
1860	380,2	24,7	0,011	9145,778		
1861	380,3	24,7	0,011	-6,0413		

Purposes (2)

- 0 Manage large amounts of raw data
- 0 **Create calculation sheets**
- 0 Create tables for reports

Microsoft Excel - Book2

File Edit View Insert Format Tools Data Window Help

Arial 10 B I U \$ % , +.0 -0.0

Formula Bar: =AVERAGE(I3:I7)

	A	B	D	E	F	G	H	I	J	K
1	Student Number		assign1	assign2	Midterm	assign3	Assign 4	Final	Mark	
2	Max Mark		10	20	50	10	10	50	100	
3	200201		9	13	45	6	9	35	76.8	
4	202110		8	15	42	7	7	46	86	
5	205345		10	16	35	5	4	33	68	
6	202347		6	15	13	8	3	44	64.6	
7	201010		7	19	44	9	8	24	67.6	
8	Average		8	15.6	35.8	7	6.2	36.4	72.6	
9	Weights		5	5	30	5	5	50	100	
10										
11										

Sheet1 / Sheet2 / Sheet3 /

Ready

0 Manage **Purposes (3)**

0 Create calculation sheets

0 **Create tables for reports**

adamfox711.wordpress.com

0

When Should Mr. Fox Drink His Coffee?		
Quantitative Data		Qualitative Data
Time (Minutes)	Temperature (Celsius)	Drinkability
0	80	scalding
1	75	painful tongue burn
1.5	70	mild tongue burn
5.5	65	blow on it and take a small sip
8	60	perfect to sip
13	55	perfect to gulp
20	50	good not great
26	45	pretty warm and passable
33	40	warm and slightly unpleasant
41	35	cool and uninviting

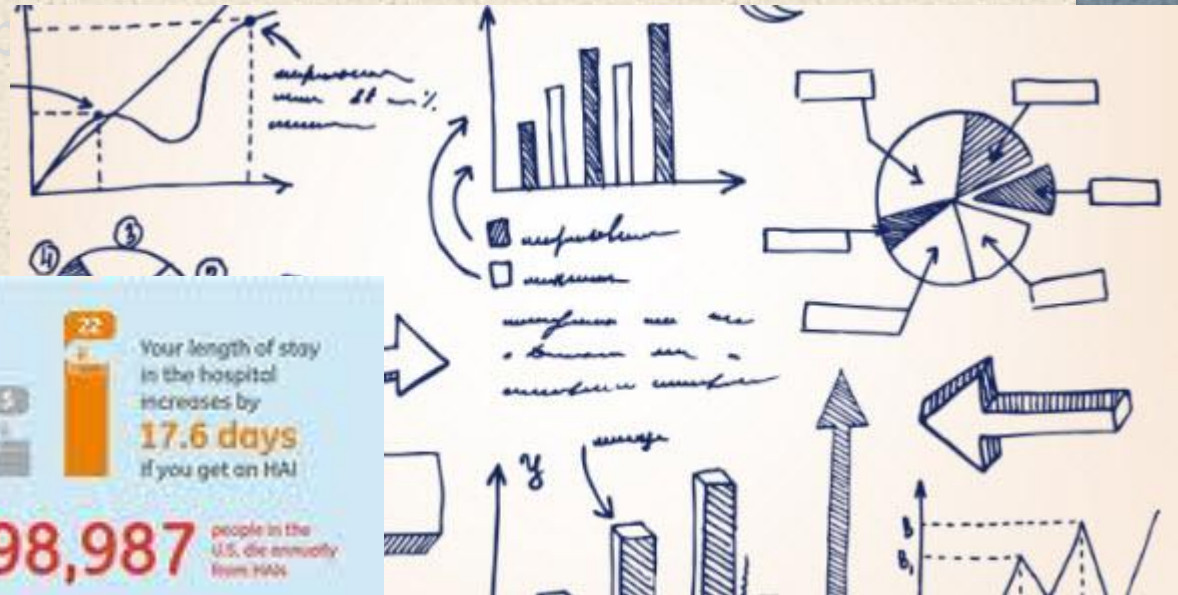
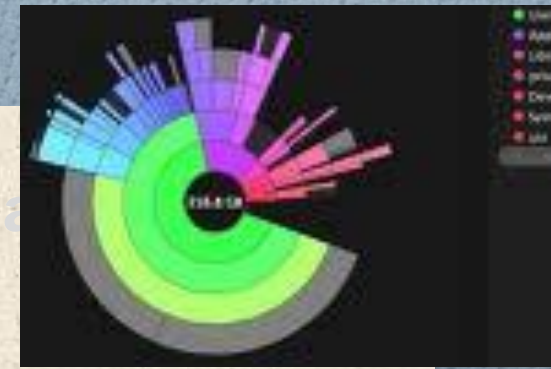
0 Manage large amounts of raw data

0 Create code for data processing

0 Create tables for reports

0 Create graphs for data visualization

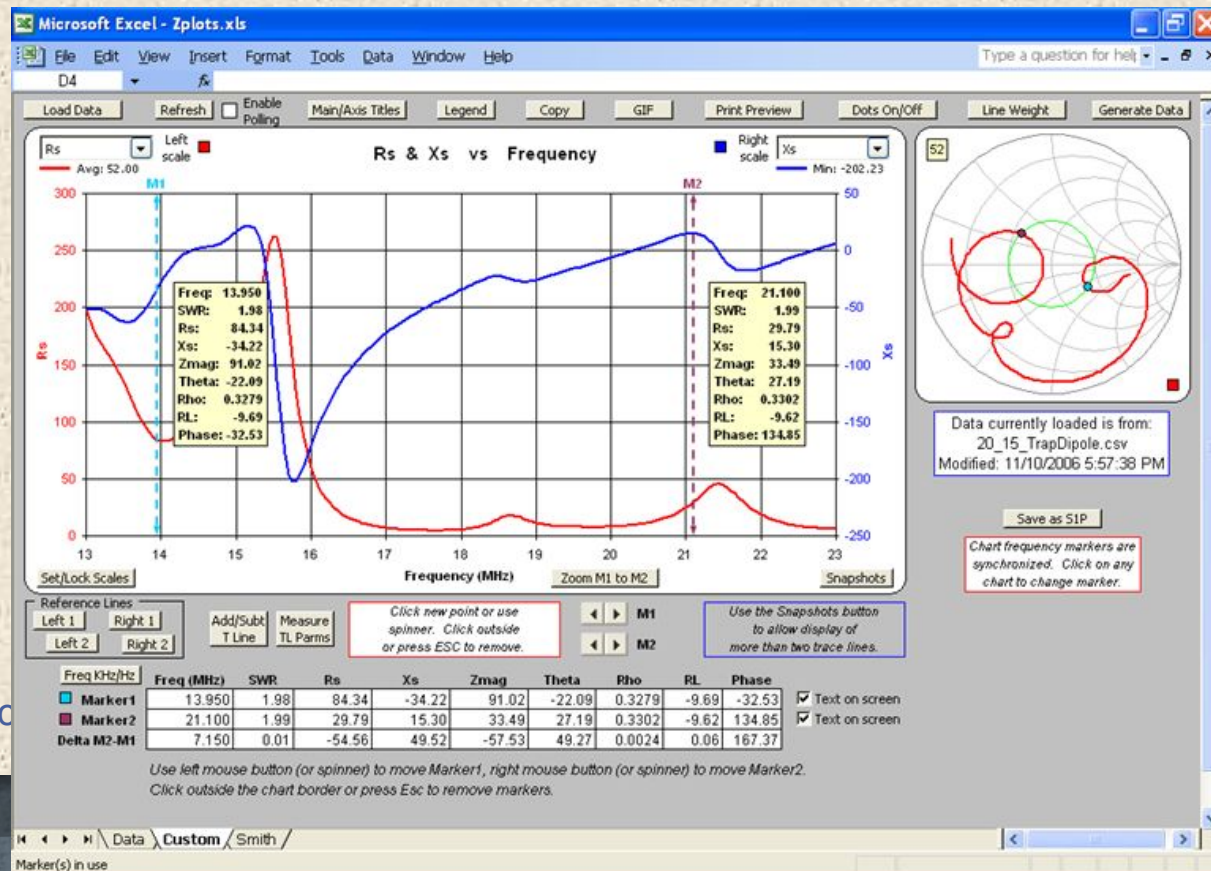
0 Create app



11/7/2014

6

- 0 Manage large amounts of raw data
 - 0 Create calculation sheets
 - 0 Create applications/tools
 - 0 Create graphs for data visualization
- # Purposes (5)



General Anatomy

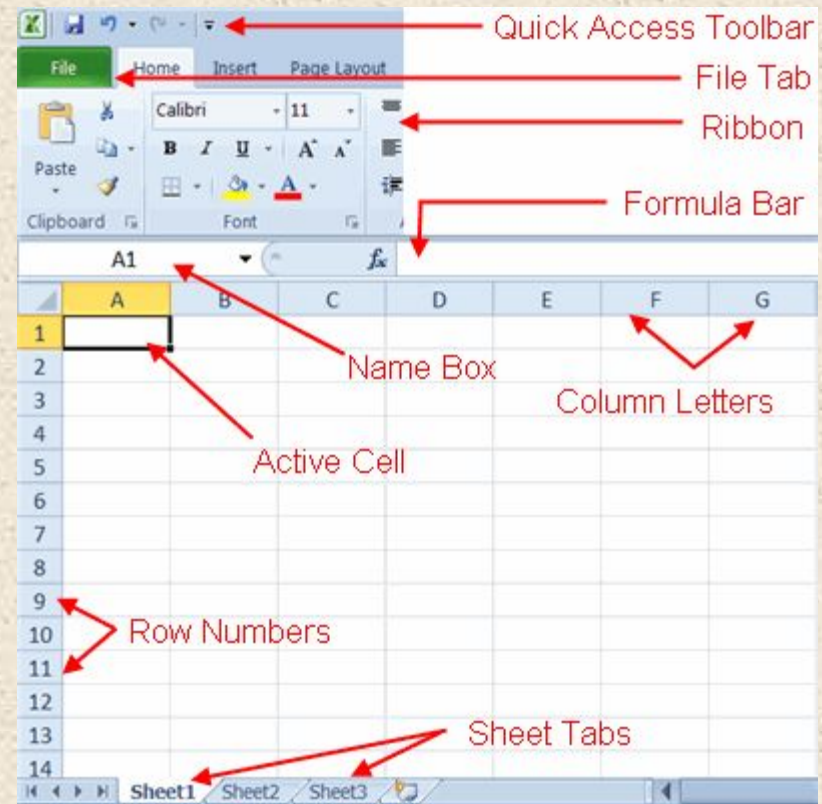
Similar to other MS applications

Data table is composed of rows (1,2,..) and columns (A,B,...)

Each cell is referred to by its corresponding column and row label numbers (e.g. A1, C23, etc.)

Clipboard compliant (cut, copy, paste, etc. works)

Selection of more than one cell is possible.

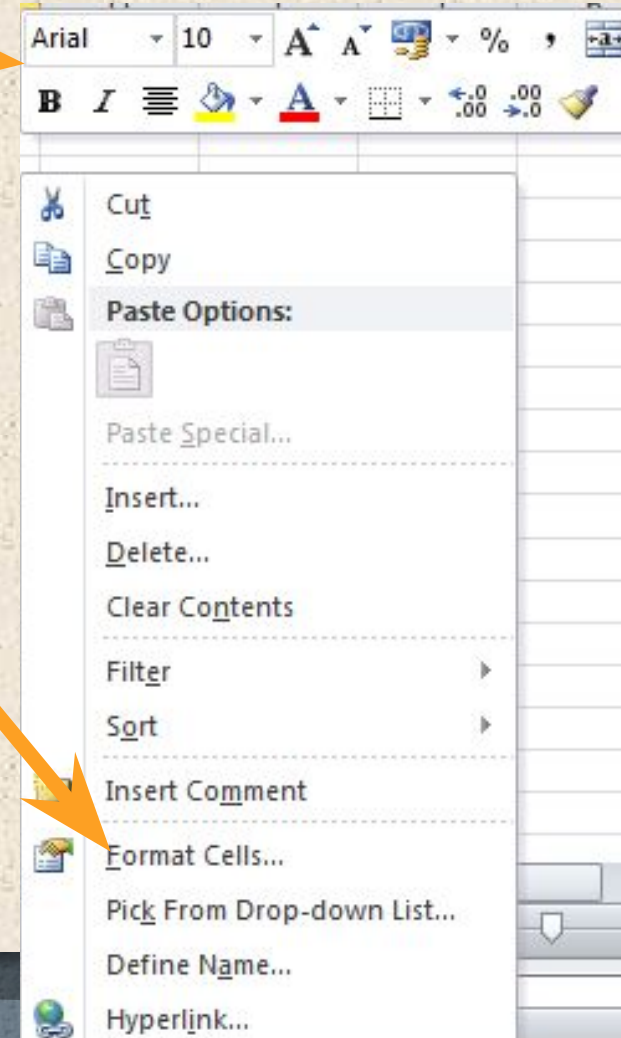


Formatting

QUICKY

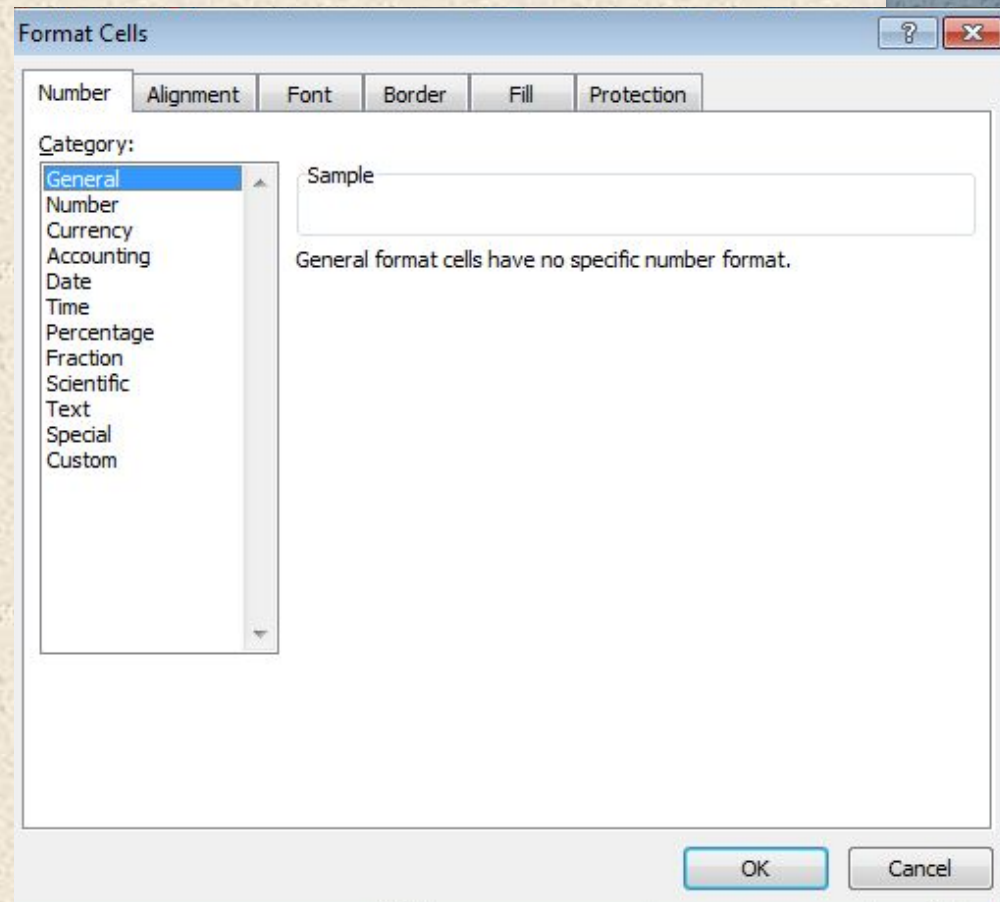
- 0 right-clicking on selected cell(s) in the data area brings the menu on the right
- 0 Cell format is adjusted by selecting “Format Cells..”
- 0 The formatting changes apply only to those cells that are currently selected.

Introduction to Mechanical Engineering



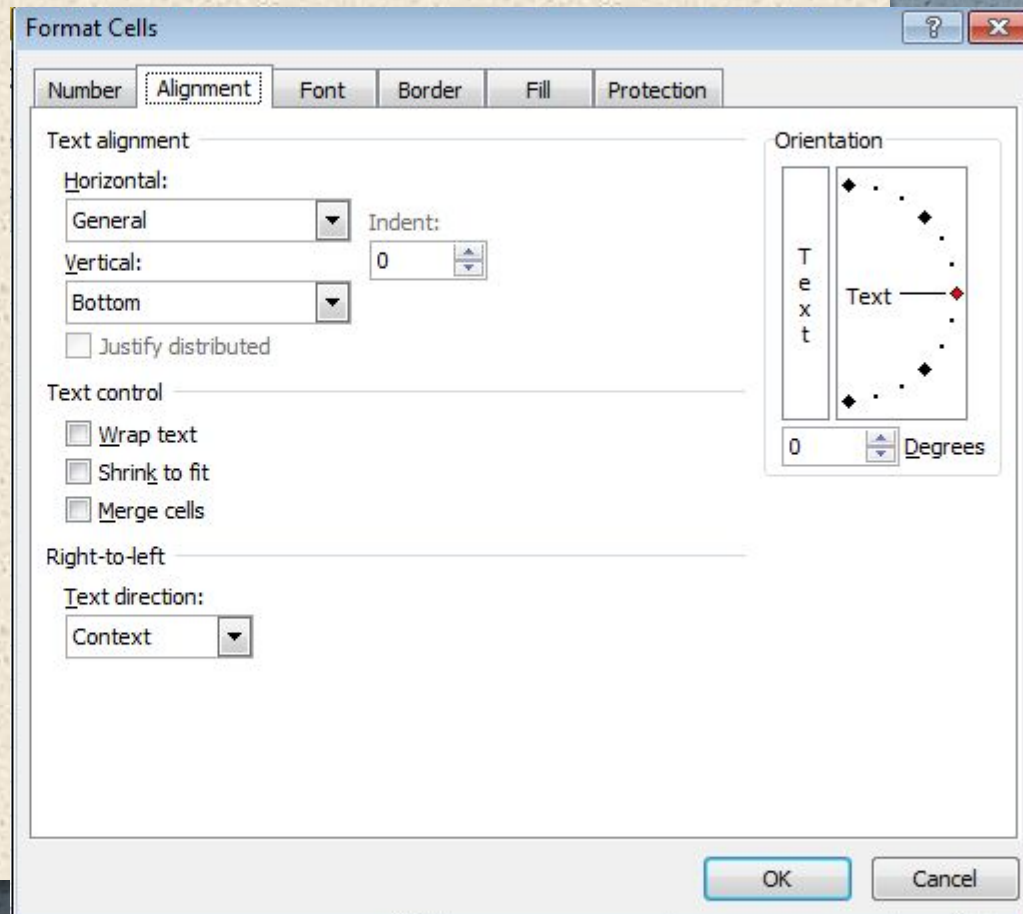
Type(cell)

- o The data stored in the cell may belong to a variety of types (number, text, date, currency, etc.)



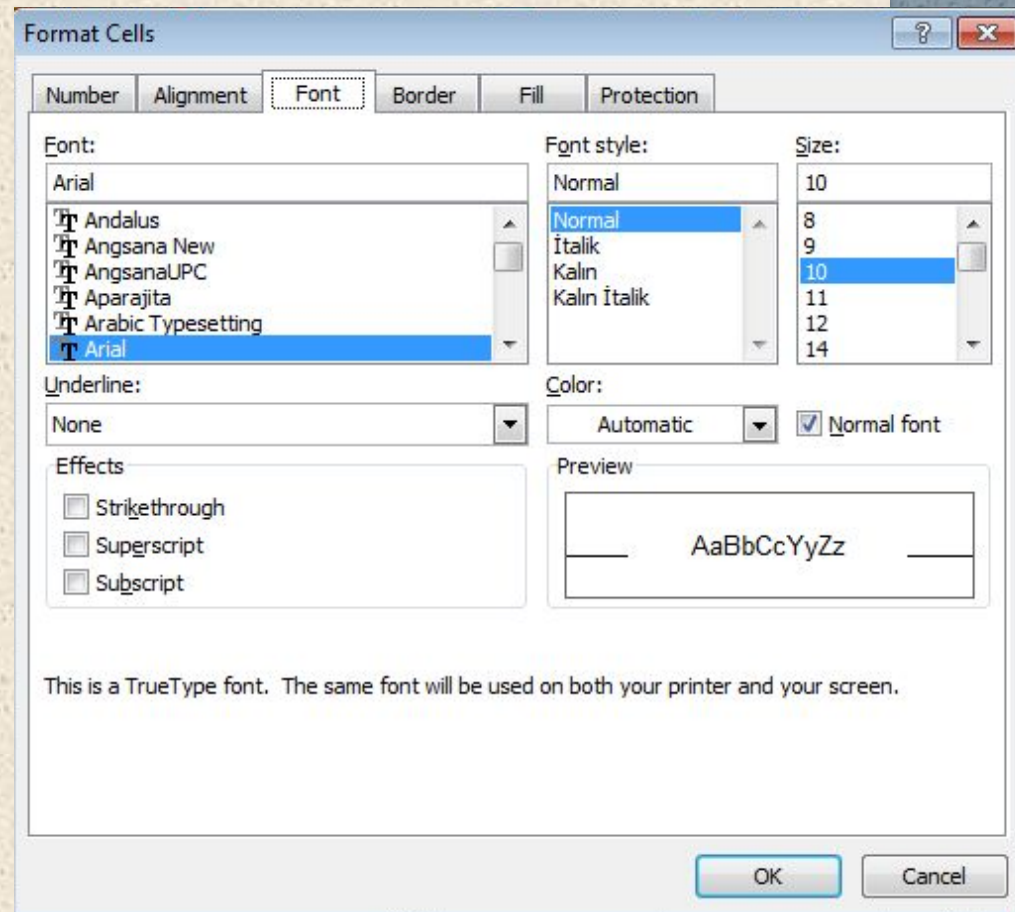
Alignment(cell)

- 0 The alignment and other related aspects can be manipulated through the “Alignment” tab of the “Format Cells” dialog.



Font(cell)

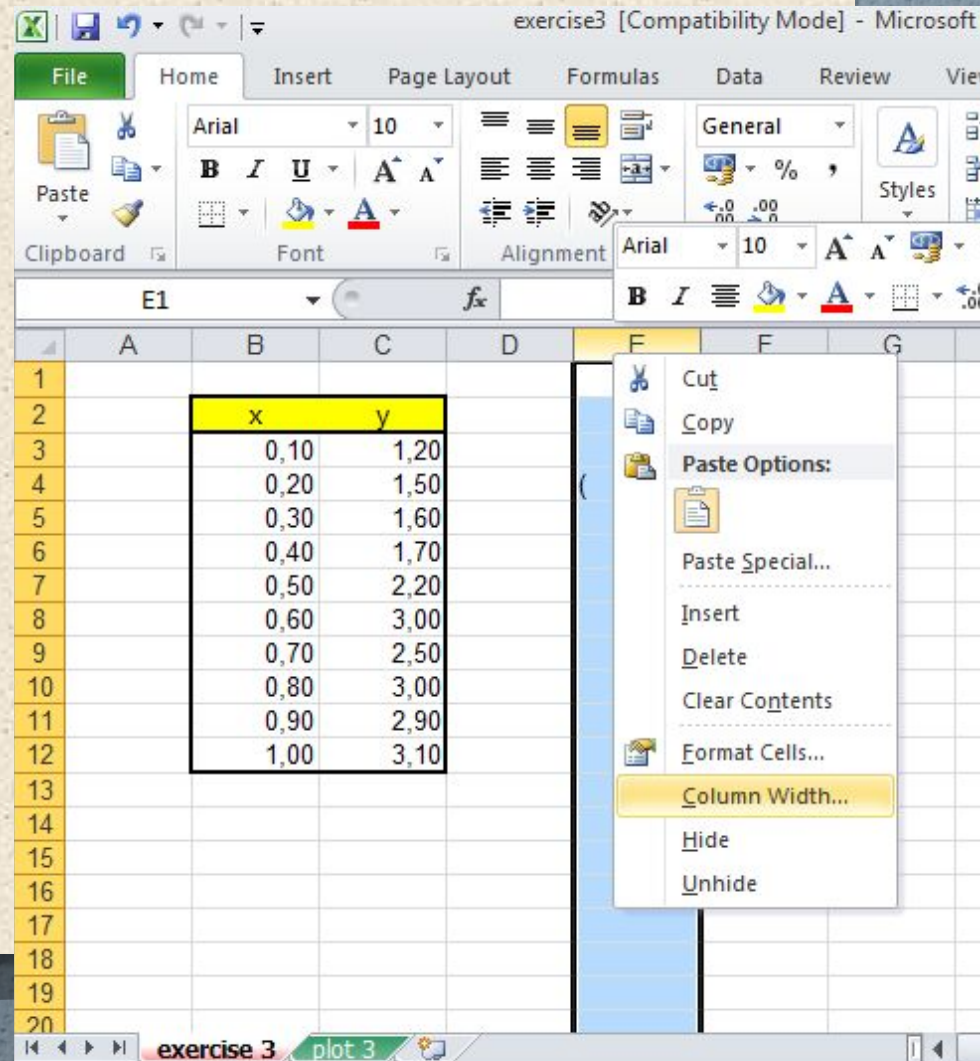
- o The font used in the cell, the cell bordering, fill properties can be accessed in a similar manner using the “Format Cells” dialog.



Size(cell)

- Cell size are controlled by its parent column and row sizes.
- Column width can be changed by right-clicking on the respective column and selecting the “column width” command.

Introduction to Mechanical Engineering



Edit, insert, format in the data area

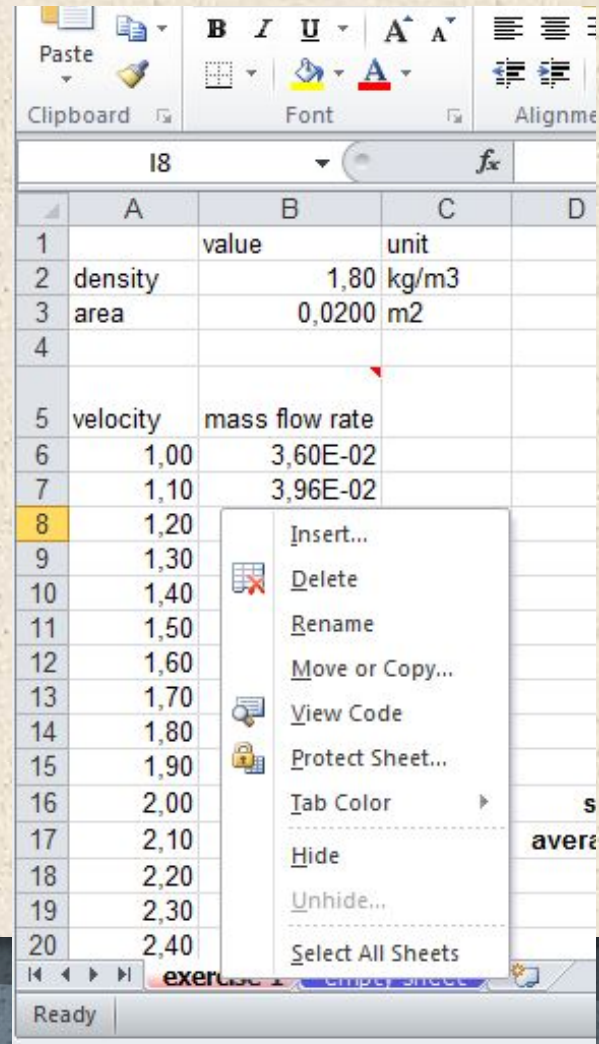
- New rows and columns can be inserted by selecting the respective row or column (left-click on the label), and pressing CTRL+, CTRL—
- Data can be manually inserted by typing in the cell area.
- Entering “=” as the first typed character initiates a formula.
- Cell label (B8, etc.) can be used as a reference to the value in that cell, anywhere else.

Exercise 1

	A	B	C	D	E	F	G
1		column width 15 pt					
2							
3		The height of row 6 is 50 pt.					
4				1,5000			
5		31.Tem.14		2,3456			
6		31.07.2014					
7		31.Tem					
8		7-31					
9							
10							
11							
12		Font size 16					
13							
14							
15							
16							
17							
18							

Worksheets

- The current data area is also called as a worksheet.
- There are more than one worksheets in a document.
- New worksheets can be added, existing ones may be deleted, copied, moved or renamed by right-clicking on the worksheet tab.



Importing/exporting data into excel

- Data recorded in an experiment and saved in a raw format such as (text, txt, ascii, csv, etc) can be imported into excel.
- Also excel data can be saved in one of these formats so as to enable accessibility by other programs (MATLAB, notepad, etc.)
- Try out saving the linked [Specimen_HamVeri_1.csv](#) onto your desktop and then import by clicking “Get External Data” from the main “Data” tab.

Sorting Data

- 0 Advanced data sorting functions are available in the “Sort & Filter” group under the main “Data” tab.
- 0 Download the linked [classlist.xlsx](#) file onto your desktop and see how the sorting and filtering functions work.

Exercise 2

Extract the following data from the classlist.xlsx file:

- Number of students whose last name start with letter B
- Number of students from the civil, chemical and electrical engineering departments
- Number of students whose student numbers start with 2010, 2011 and 2012
- Number of students who have preferred a topic in the areas of: Materials and Manufacturing, Robotics, Solids and Structures, and Thermo-fluids

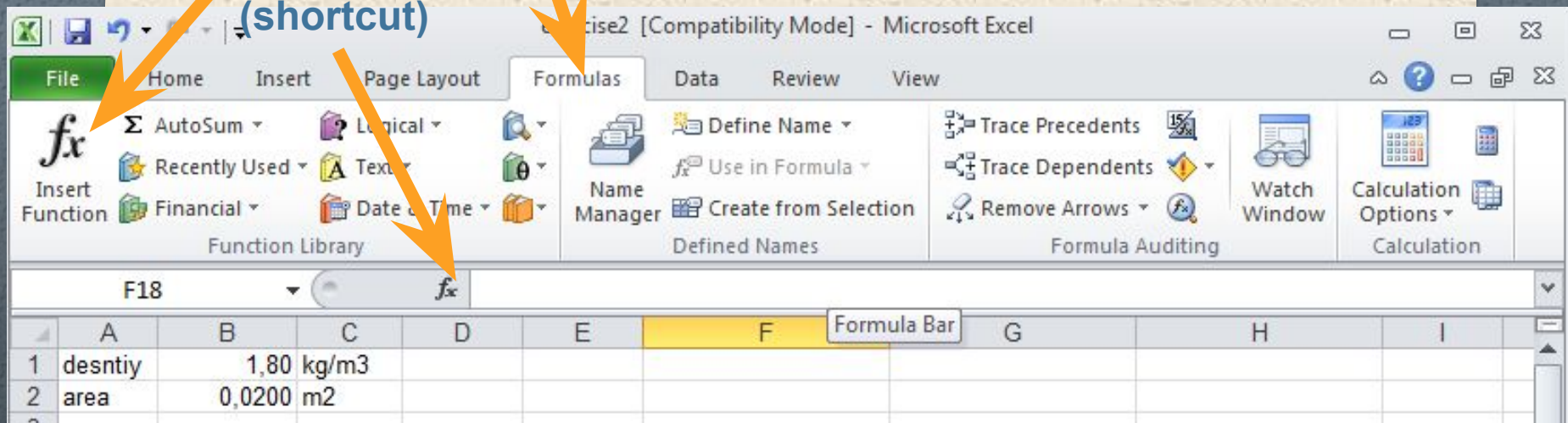
Using formulae (1)

- Formulas can be entered into a cell by typing “=” as the first character.
- The formula bar appears by clicking on the Formula tab.
- For example, typing =3+3, displays 6.

Insert function

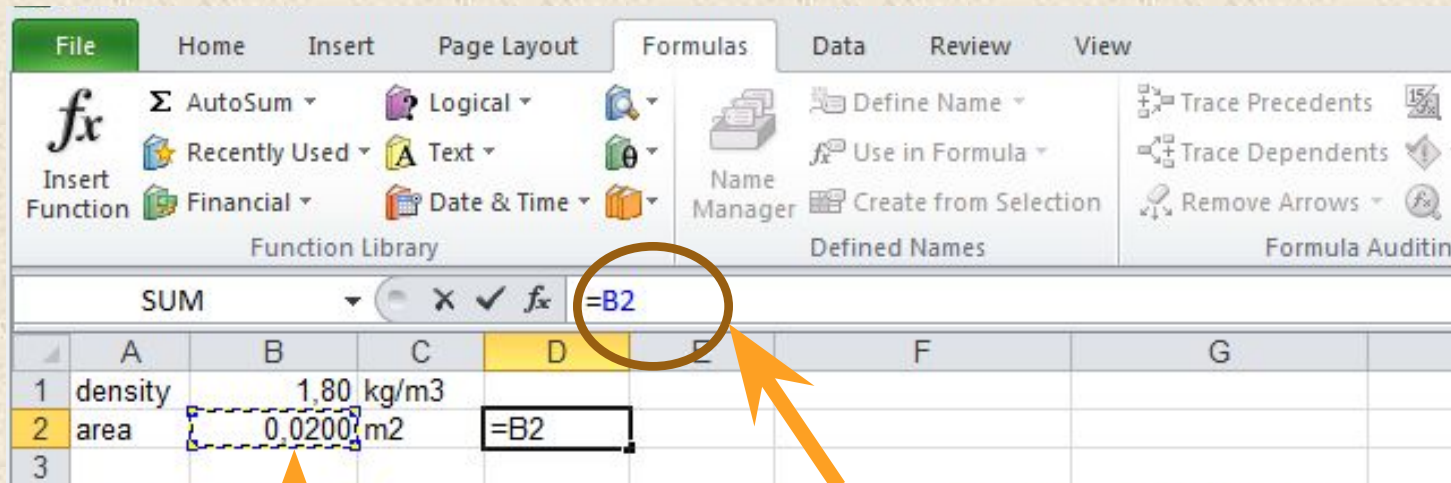
Formula tab

(shortcut)



Using formulae (2)

- When = is typed and the formula context is active, selecting any other cell or group of cells bring their names **into** the formula.

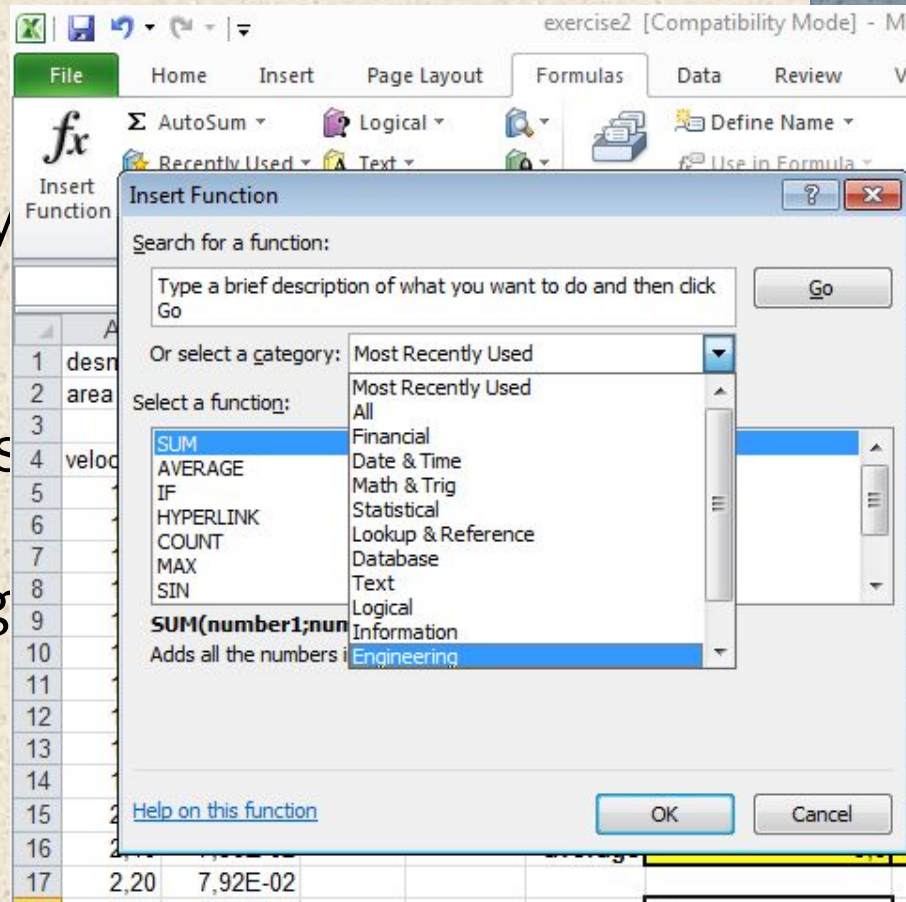


Click to select

Brings cell reference into formula

Using formulae (3)

- o A large set of built-in formulae are available
- o A formula can be used by typing its name (e.g. SIN, COS, etc) followed by a paranthesis that contains the reference cell to be used as its argument (e.g B8), so it's like:
= SIN(B8)



Using formulae (4)

- 0 AUTO-FILL is a convenient feature for filling an array of cells with the same formula. Create a list of arguments in an array form on the left column.
- 0 Enter the desired formula to the right of the topmost argument.
(e.g. = SIN(B1))
- 0 Click on the cross at the lower right corner of this cell and drag all the way down adjacent to the last argument cell.
- 0 The auto-fill feature automatically updates the reference argument according to the fill direction, i.e. B1 becomes C1 going down.

Using formulae (5)

- o In order to prevent Excel from AUTO-UPDATING the reference cell during a drag-copy operation, then the cell name (e.g. B2) should be written with a dollar sign (\$) in front of the column and row labels (i.e. as \$B\$2)

Exercise 3 (1)

The mass flow rate, $\dot{m}$, of fluid with density, ρ , flowing in a pipe with a uniform velocity, v , can be found as:

$$\dot{m} = \rho \cdot v \cdot \text{area}$$

Given: $\rho = 1.80 \text{ kg/m}^3$, $\text{area} = 0.02 \text{ m}^2$, calculate the mass flow rate for velocities ranging from 1.0 to 3.0 m/s, in increments of 0.1.

Exercise 3 (2)

	A	B	C	D	E	F	G	H
1	density	1,80	kg/m3					
2	area	0,0200	m2					
3								
4	velocity	mass flow rate						
5	1,00	3,60E-02						
6	1,10							
7	1,20							
8	1,30							
9	1,40							
10	1,50							
11	1,60							
12	1,70							
13	1,80							
14	1,90							
15	2,00							
16	2,10							
17	2,20							
18	2,30							
19	2,40							
20	2,50							
21	2,60							
22	2,70							
23	2,80							
24	2,90							

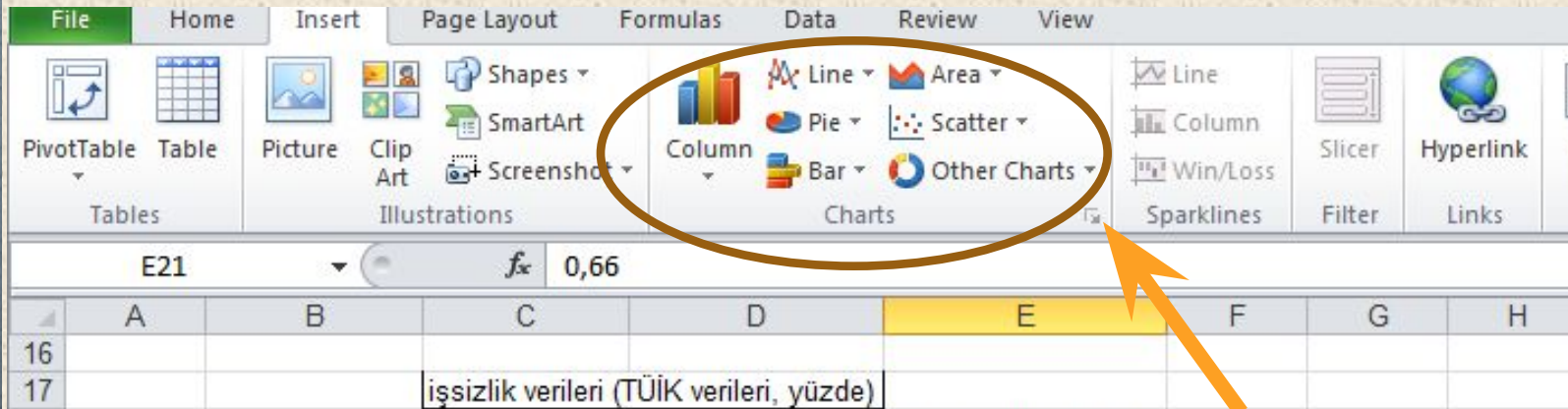
radius (m)	circumference (m)	area (m2)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
sum	55	
average	5,5	

FILL

Use the built-in SUM() and AVERAGE() functions.

Working with Charts

○ Charts are located under the “insert” tab.



Expander button

○ Click on the expander button on the lower right to get more about charts.

Chart types

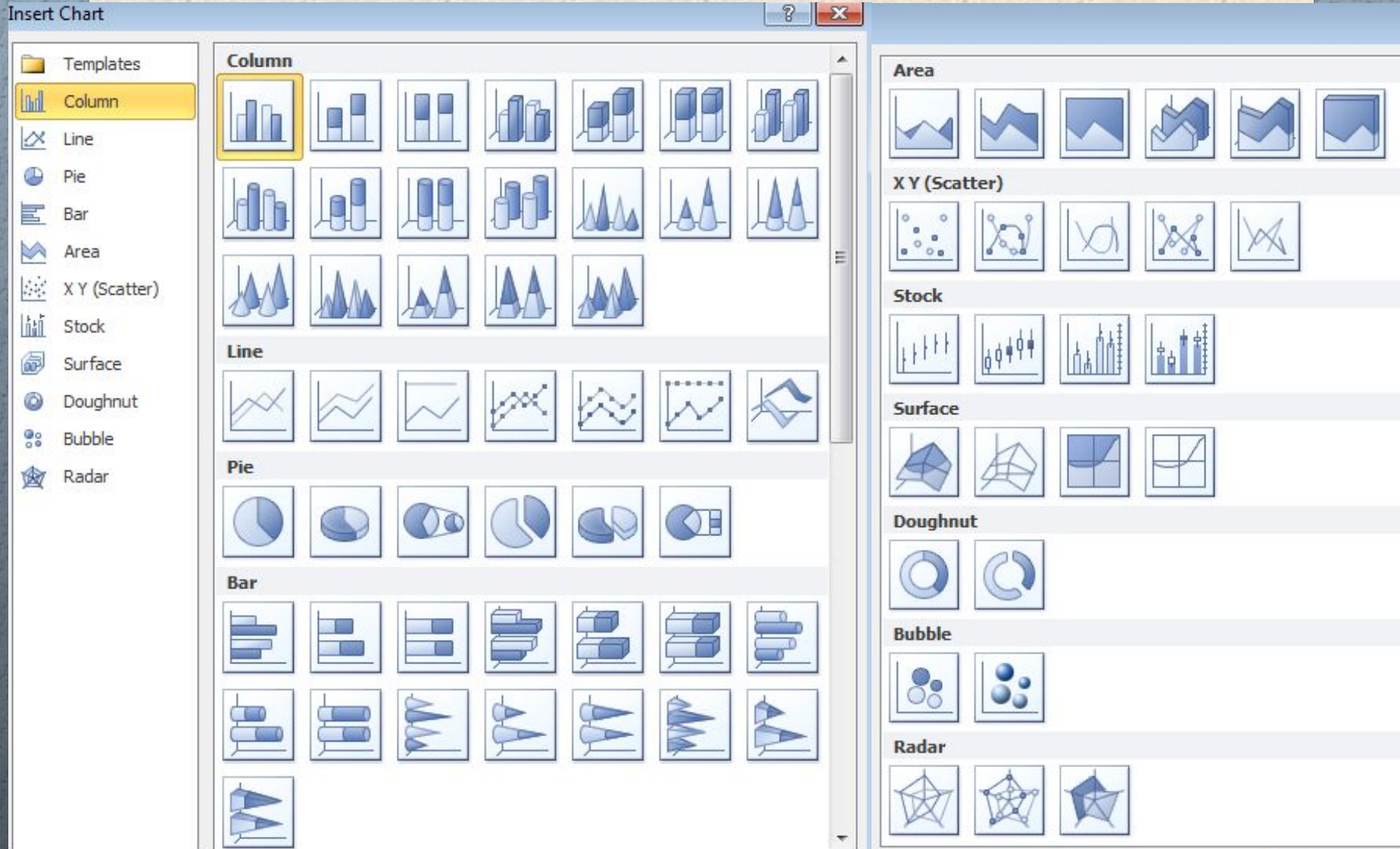
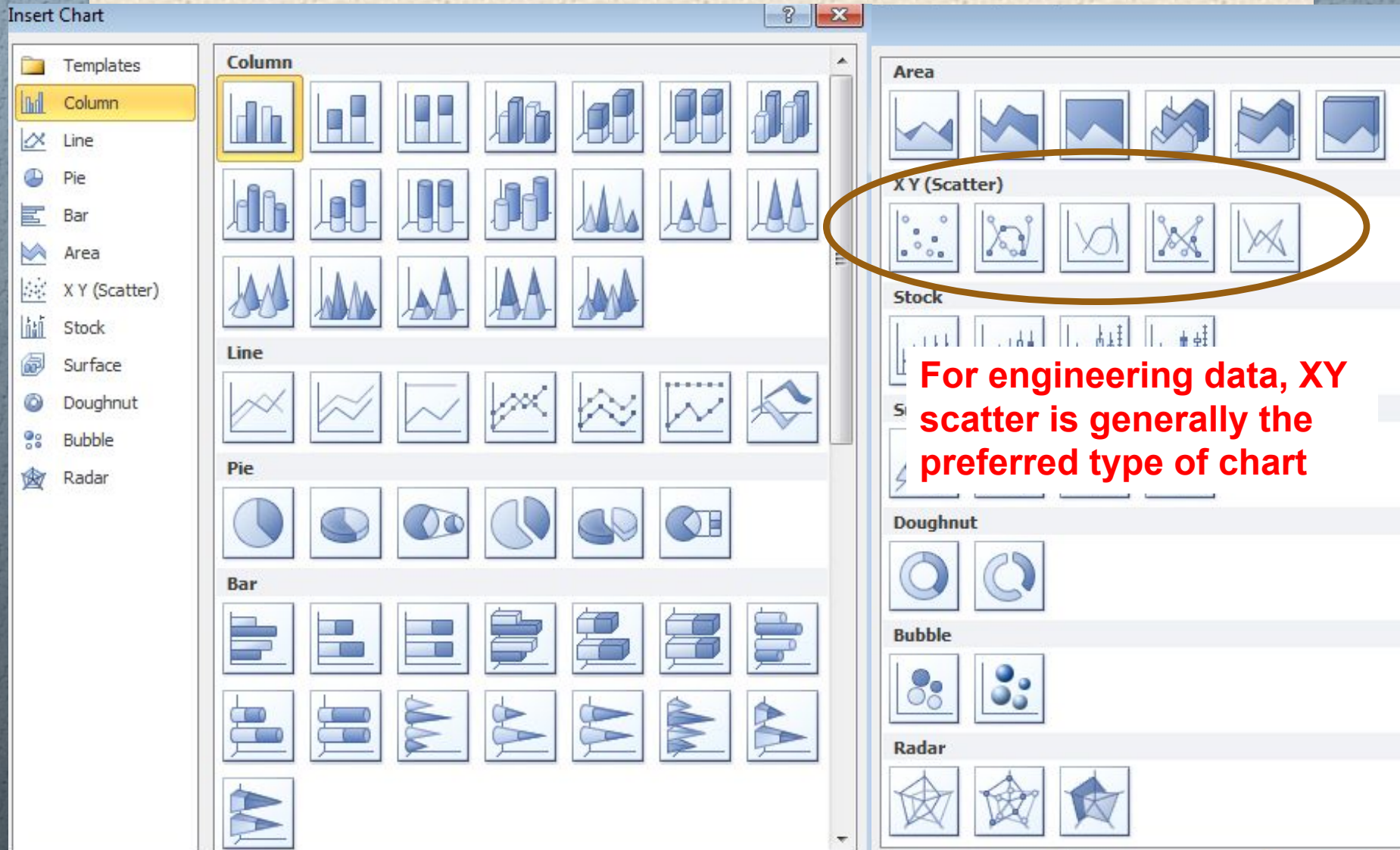


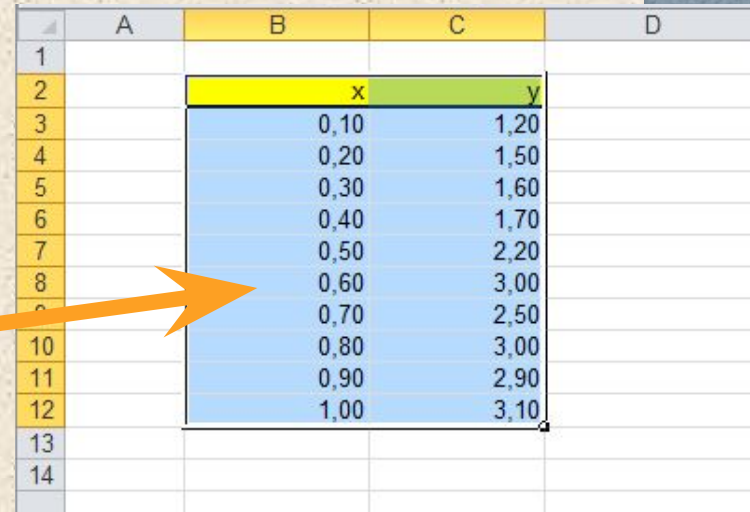
Chart types



XY Scatter Charts

- Charts are created by selecting the data range (in row or column format).

Select data range by clicking and dragging mouse



An Excel spreadsheet with columns A, B, C, and D. Row 2 is highlighted in yellow and contains 'x' in column B and 'y' in column C. Rows 3 through 12 are highlighted in blue and contain numerical data. An orange arrow points from the text 'Select data range by clicking and dragging mouse' to the blue highlighted data range.

	A	B	C	D
1				
2		x	y	
3		0,10	1,20	
4		0,20	1,50	
5		0,30	1,60	
6		0,40	1,70	
7		0,50	2,20	
8		0,60	3,00	
9		0,70	2,50	
10		0,80	3,00	
11		0,90	2,90	
12		1,00	3,10	
13				
14				

- Then clicking on the appropriate chart type from the Insert □ Charts group, brings the chart.

Introduction to Mechanical Engineering

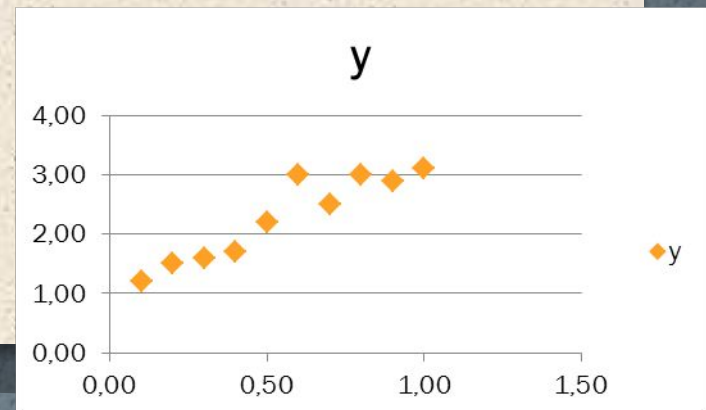


Chart objects

- Charts have various objects whose properties and formats may be manipulated as desired.

Axis label

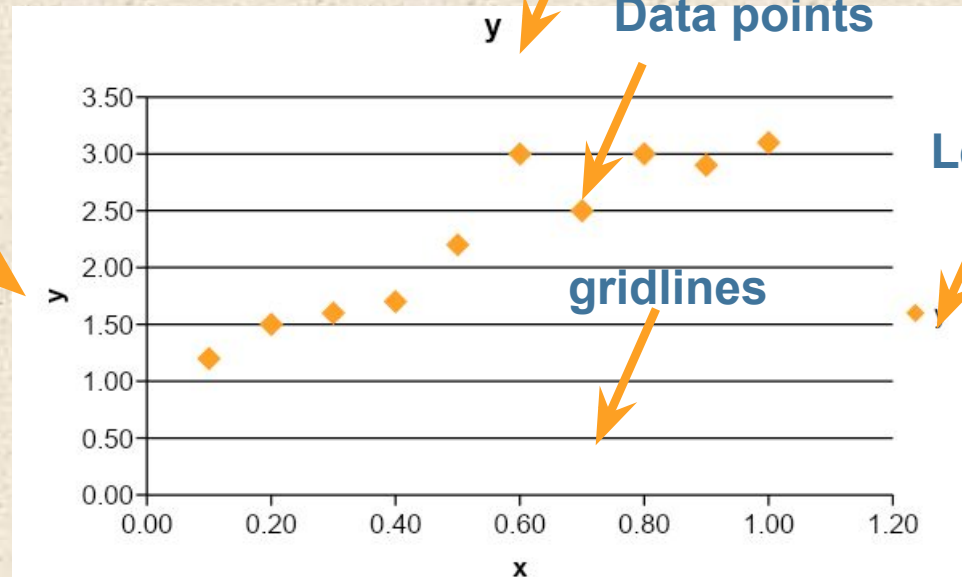


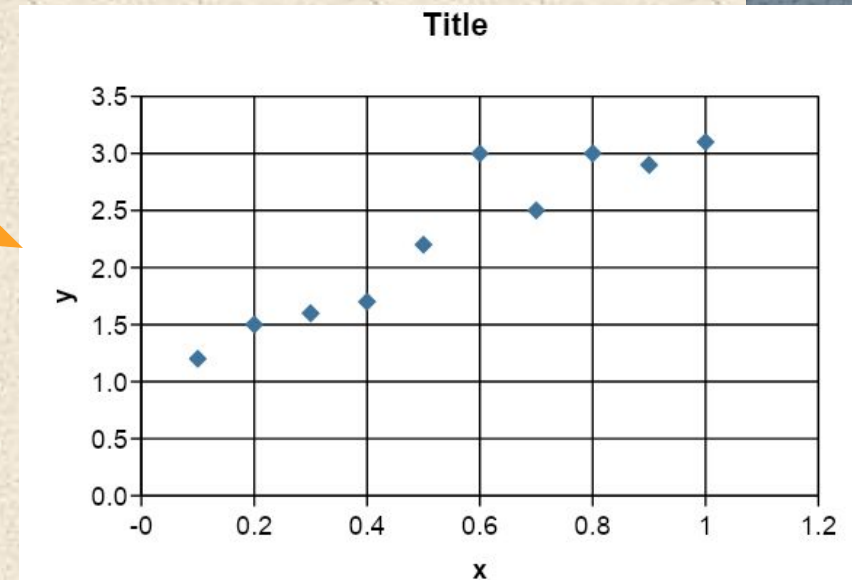
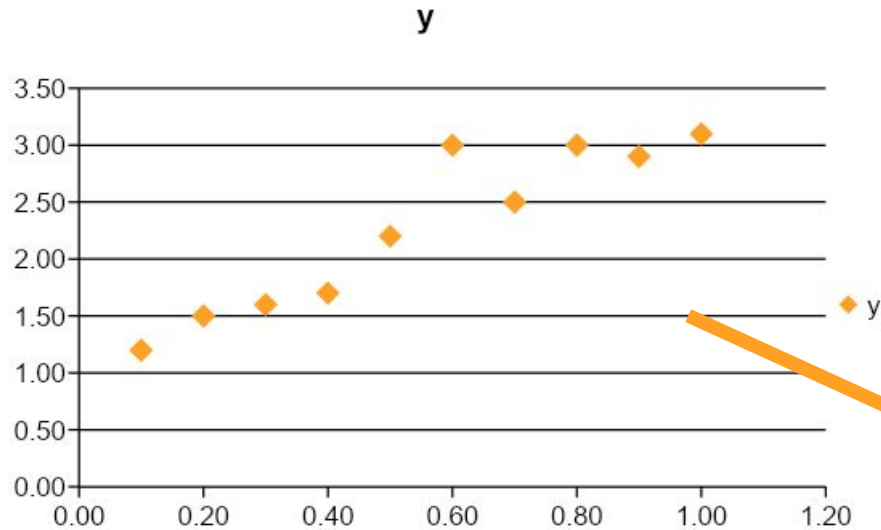
Chart title

Data points

gridlines

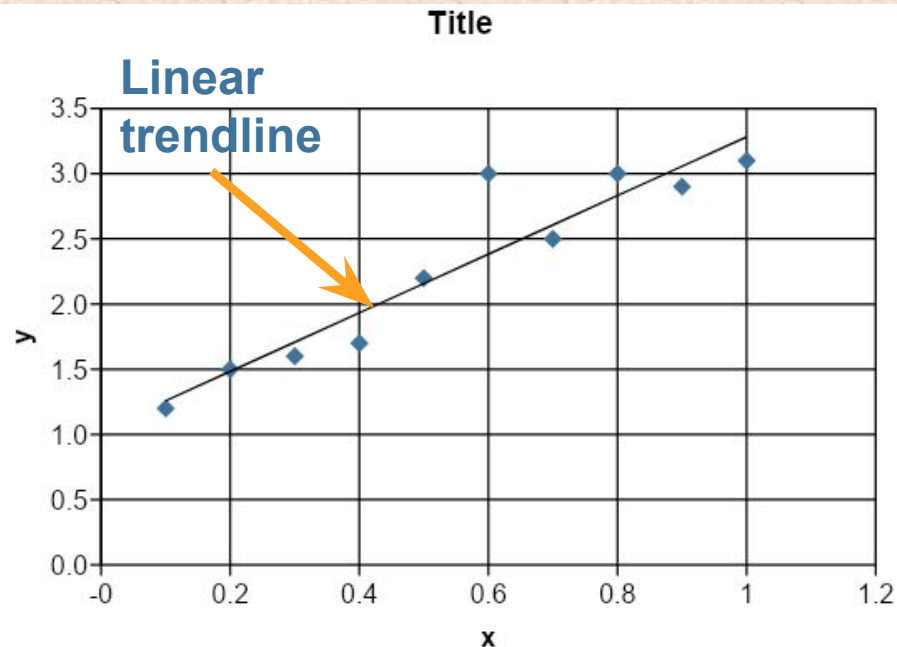
Legend

Manipulating chart objects



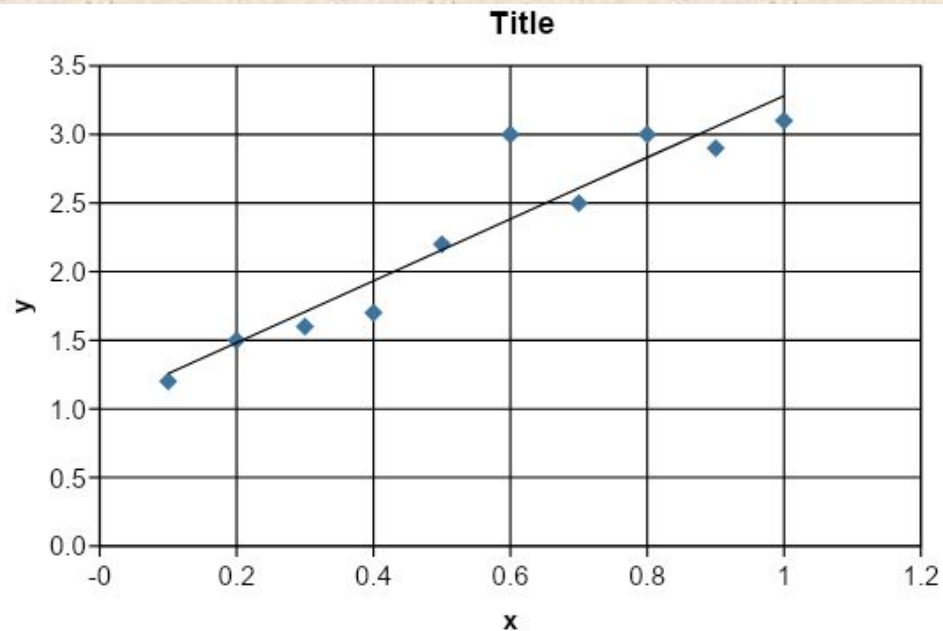
Trendline objects

- Trendlines are used to depict the variation of data, by way of fitting analytical functions onto data points.
- There are various kinds of analytical functions used: linear, quadratics, exponential, etc.



Exercise 4

- 0 Download [exercise4.xls](#)
- 0 Create an XY scatter chart from the data.
- 0 Format it to look like this.
- 0 Add a trendline



Exercise 5

- 0 Download [exercise5.xls](#)
- 0 Create a column chart of unemployment figures in 2013
- 0 Create a line chart of inflation figures in the same year.
- 0 Format the charts so as to superimpose them as a single chart, to point out correlation (if any) between the two economic indicators.